

ORLANDO INTL, FL, USA

WMO: 722050

Lat: 28.434N

Lon: 81.325W

Elev: 27

StdP: 101.00

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 12815

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.6	5.8	-6.3	2.2	8.8	-3.2	2.9	10.2	10.6	16.2	9.3	16.3	3.5	350	0.379

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.8	34.3	24.8	33.5	24.6	32.8	24.4	26.5	30.6	26.1	30.2	25.7	29.8	3.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.5	20.8	27.6	25.1	20.3	27.2	24.7	19.8	27.0	82.2	30.7	80.6	30.3	79.1	30.0	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.1	8.1	7.4	DB	0.0	35.8	2.2	0.9	-1.6	36.5	-2.9	37.0	-4.2	37.5	-5.8	38.2	(4)
			WB	-2.1	27.4	2.1	0.7	-3.5	27.9	-4.7	28.2	-5.9	28.6	-7.4	29.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	22.8	15.8	17.7	19.7	22.5	25.3	27.3	28.1	28.1	27.2	24.2	20.2	17.7	(6)
(7)		DBStd	5.24	4.53	4.27	3.68	2.74	1.96	1.44	1.14	1.11	1.23	2.86	3.43	4.42	(7)
(8)		HDD10.0	15	8	3	1	0	0	0	0	0	0	0	0	3	(8)
(9)		HDD18.3	284	102	57	30	4	0	0	0	0	0	3	22	66	(9)
(10)		CDD10.0	4704	189	218	302	374	475	519	560	562	516	442	305	243	(10)
(11)		CDD18.3	1933	24	40	73	128	217	269	302	303	266	186	77	47	(11)
(12)		CDH23.3	16326	141	262	526	1031	1911	2402	2879	2885	2300	1343	417	231	(12)
(13)	CDH26.7	6008	9	43	111	318	776	1023	1221	1167	853	394	67	25	(13)	
(14)	Wind	WSAvg	3.4	3.6	3.7	3.9	3.8	3.5	3.0	2.6	2.7	3.1	3.5	3.5	3.4	(14)
(15)	Precipitation	PrecAvg	1241	62	63	84	52	93	189	187	175	155	70	49	57	(15)
(16)		PrecMax	1724	184	222	323	231	370	442	464	403	358	259	261	321	(16)
(17)		PrecMin	772	3	4	0	0	9	40	66	72	42	0	1	5	(17)
(18)		PrecStd	214	44	49	71	47	68	81	78	71	66	57	54	61	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.9	29.5	30.6	32.7	34.2	35.7	35.1	34.9	33.7	32.4	30.1	28.7	(19)
MCWB			20.7	21.0	21.4	22.1	22.7	24.5	25.3	25.4	24.5	24.4	22.9	21.9	(20)	
2%		DB	26.5	28.0	29.2	31.1	33.0	34.2	34.1	33.9	32.8	31.2	28.5	27.3	(21)	
		MCWB	19.9	20.3	20.5	21.6	22.5	24.6	25.1	25.2	24.5	23.9	22.0	21.2	(22)	
5%		DB	25.0	26.7	27.9	29.9	32.0	33.0	33.2	33.0	32.0	30.1	27.2	26.1	(23)	
		MCWB	19.2	19.6	19.6	20.9	22.3	24.3	25.0	25.1	24.3	23.3	21.0	20.5	(24)	
10%		DB	23.2	24.9	26.5	28.6	30.8	31.8	32.3	32.1	31.1	29.0	25.8	24.4	(25)	
	MCWB	18.5	18.8	19.0	20.2	22.1	24.2	25.0	25.1	24.3	22.7	20.1	19.8	(26)		
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.3	22.4	23.4	24.1	25.2	26.7	26.9	26.6	26.0	24.2	23.1	(27)	
MCD B			25.4	26.3	27.3	28.8	30.2	31.3	31.0	31.1	30.1	29.0	27.3	26.6	(28)	
2%		WB	21.1	21.6	22.2	23.2	24.4	25.8	26.4	26.5	25.8	25.3	23.4	22.3	(29)	
		MCD B	24.3	25.4	26.2	27.8	29.2	30.4	30.6	30.7	29.4	28.4	26.3	25.3	(30)	
5%		WB	20.3	20.8	21.4	22.4	23.9	25.4	25.9	26.1	25.6	24.8	22.6	21.6	(31)	
		MCD B	23.4	24.6	25.4	27.0	28.6	30.1	30.2	30.2	29.2	27.9	25.4	24.3	(32)	
10%		WB	19.3	20.0	20.5	21.7	23.3	25.0	25.6	25.7	25.2	24.2	21.6	20.7	(33)	
	MCD B	22.3	23.7	24.5	26.3	27.9	29.5	29.9	29.7	28.8	27.2	24.4	23.4	(34)		
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.3	11.4	11.6	11.5	11.0	9.5	9.2	8.8	8.3	9.1	10.2	10.4	(35)
MCD BR			11.9	12.2	12.4	12.4	12.2	11.0	10.1	9.8	9.2	9.3	10.4	10.5	(36)	
MCWBR			6.0	5.2	4.8	4.3	3.6	3.3	3.0	2.9	2.8	3.3	4.5	5.1	(37)	
5% WB		MCD BR	10.6	10.7	10.6	11.0	10.4	9.4	9.2	8.8	8.2	7.8	8.8	9.2	(38)	
		MCWBR	6.2	5.3	5.1	4.6	3.6	3.4	3.2	3.0	3.0	3.2	4.6	5.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.339	0.353	0.373	0.391	0.425	0.454	0.475	0.463	0.422	0.397	0.358	0.346	(40)	
taud		2.524	2.496	2.429	2.357	2.309	2.300	2.239	2.289	2.398	2.431	2.495	2.526	(41)		
Ebn at Noon		904	919	917	905	870	839	821	829	857	859	874	878	(42)		
Edh at Noon		90	100	113	125	132	132	140	133	115	105	92	86	(43)		
(44)	All-Sky Solar Radiation	RadAvg	3.43	4.07	5.14	6.08	6.30	5.66	5.66	5.39	4.85	4.42	3.63	3.13	(44)	
(45)		RadStd	0.25	0.26	0.38	0.30	0.43	0.52	0.34	0.36	0.27	0.30	0.19	0.24	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	-45	+135	+100	(46)
(47)	Regional (41 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	-5	-47	+158	+113	(47)

Nomenclature: See separate page